

What it costs to deploy IFQM and hand it to the client

A production hosting and cost plan for a multi-tenant

React + Node/Express + MySQL platform sized for **5,000–10,000**

employees per organisation across several tenants — built to run without lag and to survive an enterprise infosec review.

Currency · INR (₹)

Region · India (data residency)

FX assumed · ₹85 / USD

Prepared · Jul 2026

THE SHORT ANSWER

Vercel for the frontend — yes. Render/Vercel for the backend — no. Host the API on a real server in an India region with a managed database.

This app is a **stateful monolith**: it runs background import jobs inside the Node process, keeps rate-limit counters in memory, spawns CPU worker-threads for password hashing, and writes uploaded attachments to local disk. Those four traits are exactly what serverless platforms (Vercel functions) and ephemeral-disk hosts break. It wants a server it can keep, not a function that comes and goes.

RECOMMENDED HOST	FRONTEND	MONTHLY · PRODUCTION
Cloud VM + managed MySQL	Vercel / Cloudflare (static) · Free	₹28k–38k / mo
GEMINI AI SCORING	DOMAIN	SSL / HTTPS
≈ ₹200–1,500 / mo	₹800–1,500 / yr	Free (Let's Encrypt / ACM)

! Before this goes to a TATA / L&T / TVS-tier client, budget a **pre-handover hardening pass** and (ideally) an external security test. The app's own runbook lists seven known gaps — some matter a lot when the data is real Indian employee PII under the DPDP Act. See §6.

01 Why Vercel + Render is the wrong shape for the backend

Vercel and Render are excellent for stateless apps. IFQM's backend is stateful in four specific, load-bearing ways — each one is documented in the code, not a guess.

WHAT THE CODE DOES	WHERE	WHY SERVERLESS / EPHEMERAL HOSTING BREAKS
Background import jobs run <i>inside</i> the Node process A detached promise created by <code>runJob()</code> keeps working after the HTTP response is sent — a 10,000-employee import takes ~2 min.	<code>userImportService.js</code>	A serverless function is killed the instant it returns response. The import would silently die half-way.
Uploaded attachments are written to local disk <code>backend/uploads/<org>/</code> — the runbook explicitly says a DB-only backup loses them.	<code>uploadService.js</code>	Vercel/Render web instances have an ephemeral filesystem — every deploy or restart wipes the file. Needs a persistent disk or object storage.
Rate limiting is stored in process memory <code>express-rate-limit</code> default store — the runbook flags this in §8.6.	<code>rateLimiter.js</code>	Scale to more than one instance and each one keeps its own counter, so the effective limit multiplies. Needs a shared Redis store to scale out.
Password hashing uses CPU worker-threads <code>hashPool.js</code> spins up to 8 threads to hash thousands of bcrypt passwords in parallel.	<code>hashPool.js</code>	Wants real, multi-core CPU it can hold for the duration — not a throttled, short-lived function invocation.
Database is MySQL, addressed with raw SQL <code>mysql2</code> , one master registry + one schema per tenant.	<code>tenant.js</code> / <code>master.js</code>	Render offers managed <i>Postgres</i> , not MySQL. On Render you must bring your own MySQL — which removes most of its convenience.

- The **frontend is a different story**. It's a static Vite build (`frontend/dist`) — put it on Vercel or Cloudflare Pages for free, or just serve it from the same server via Nginx as the runbook already describes. Either is fine.

02 Hosting options, compared

Ranked for this specific app and an India-based enterprise rollout. "Fit" weighs the stateful architecture, MySQL requirement, DPDP data-residency, and how a client's infosec team will view it.

OPTION	BACKEND	MYSQL	FILES	VERDICT FOR IFQM
Cloud VM (India) + managed MySQL DigitalOcean Bengaluru, or AWS/Azure Mumbai. Nginx + systemd — exactly what the runbook is written for.	Ideal	Managed	Disk / S3	Recommended. Matches the architecture and the existing deploy runbook 1:1; full control for compliance.
AWS / Azure Mumbai (managed stack) EC2/ECS + RDS MySQL + S3 + ElastiCache, Multi-AZ.	Ideal	RDS	S3	Best for TATA/L&T-tier. The name enterprise infosec expects; highest cost and ops overhead.
Render (web service) Long-running Node process, persistent disk add-on.	Workable	Not offered	Paid disk	Possible, but you must host MySQL elsewhere and pin to one instance. Convenience mostly evaporates.
Vercel (functions) Serverless.	Breaks it	No	No	Frontend only. Do <i>not</i> run the API here — see §1.
Shared / cPanel hosting Traditional PHP-style hosting.	No	Bundled	Disk	Won't run a long-lived Node service or worker-threads reliably. Skip.

03 Monthly running cost — three tiers

Pick by stage, not by hope. Note: 10,000 users per tenant is the *register*, not the *concurrency*. An ideation tool sees maybe 1–5% of staff active at peak, and each request is light (JWT auth + CRUD). These tiers are sized with comfortable headroom so there is no lag.

Pilot

First tenant / proof-of-value. One org, a few thousand employees.

₹7k–9k / month

- 1× app VM (4 vCPU / 8 GB)
- Managed MySQL (2 GB)
- Object storage + CDN
- Backups + low-volume email
- Cloudflare free tier

RECOMMENDED

Production

Several tenants, 5k–10k each, "no lag" SLA. Redundant app tier + HA database.

₹28k–38k / month

- 2× app VM behind a load balancer
- Managed MySQL w/ standby (HA)
- Managed Redis (shared rate-limit)
- Object storage + CDN + backups
- Email (SES) + monitoring

Enterprise HA

AWS/Azure Mumbai, Multi-AZ, WAF. For procurement at TATA / L&T / TVS scale.

₹90k–1.5L / month

- 2–3× EC2/ECS across zones
- RDS MySQL Multi-AZ + read replica
- ElastiCache Redis (HA pair)
- S3 + CloudFront + WAF + Shield
- CloudWatch, cross-region backups

ITEMISED – PRODUCTION TIER (THE RECOMMENDED ONE)

COMPONENT	SPEC	₹ / MONTH
Application server(s)	2× 4 vCPU / 8 GB (Node behind Nginx)	9,000
Load balancer	Managed LB + TLS termination	1,100
Managed MySQL	4 vCPU / 16 GB, HA standby, auto-backup	9,500
Managed Redis	1–2 GB — shared rate-limit / cache	1,700
Object storage + CDN	~500 GB attachments, served via CDN	900
Backups & snapshots	Off-machine DB dumps + uploads archive	1,000
Transactional email	AWS SES / Brevo — notifications	1,700
Monitoring & logs	Uptime, metrics, alerting	1,300
AI scoring (Gemini)	See §4 — effectively rounding error	≈ 600
Total	Recommended production stack	≈ 26,800

Range quoted as ₹28k–38k to absorb data-transfer, a Cloudflare Pro/WAF add-on, and provider price variation. Prices are provider list rates (mid-2026) at ₹85/USD; reserved/committed-use discounts can cut the compute lines 20–40%.

04 **Gemini API — AI scoring & reasoning cost**

The app scores each idea on submit (and on demand) with `gemini-2.0-flash`, capped at 150 output tokens, and falls back to a built-in heuristic if the key is unset. The prompt is tiny. This is the cheapest line in the whole budget.

■ **Cost per idea scored**

~400 input + ~120 output tokens. At Gemini 2.0 Flash rates (≈ \$0.10 / M input, \$0.40 / M output):

≈ **₹0.008** per idea — under one paisa.

■ **At realistic volume**

Even a heavy year — say **100,000 scorings** including admin bulk-rescores across all tenants:

≈ **₹800 / year**. A single 20,000-idea bulk rescore costs about **₹170**.

■ **What to actually budget**

Set a generous ceiling of **₹200–1,500 / month** that you will almost never hit. Use the paid (pay-as-you-go) tier, not the free tier, so burst rate-limits don't throttle a bulk rescore.

■ **If you want deeper reasoning**

Swapping to a reasoning model (e.g. Gemini 2.5 Pro) raises cost ~30–50× — to roughly **₹0.25–0.50 per idea**. Still trivial: even 100k scorings ≈ ₹25k–50k *per year*. One-line change in `aiService.js`.

- ✓ Bottom line: **AI is not a cost driver**. Don't let anyone size the deal around it. If the Gemini key is ever removed, scoring silently falls back to the local heuristic at zero cost and the app keeps working.

05 Initial / one-time costs

Setup costs incurred once, at handover. Domain and SSL are trivial; the real one-time investment is the engineering to close the known gaps (§6) and, for enterprise clients, an independent security test.

ITEM	NOTES	₹ (ONE-TIME)
Domain name	.com ≈ ₹1,200/yr · .in ≈ ₹800/yr (recurring yearly, not monthly)	800–1,500
SSL / TLS certificate	Let's Encrypt or AWS ACM — auto-renewing	0
Deployment & handover engineering	Server hardening, migrations, Nginx/systemd, backups + tested restore, runbook walkthrough	40k–1.5L
Pre-handover hardening (see §6)	DPDP export/delete, Redis rate-limit, upload virus scan, security audit log, first automated tests	1.5L–3L
External security test (VAPT)	Strongly advised before a TATA/L&T-tier infosec review; they will likely ask for the report	75k–2.5L
Load / soak test	Prove "no lag" at target concurrency before go-live	included / 25k

The engineering and VAPT figures are market ranges for your planning — the domain (₹≈1,200/yr) and SSL (free) are the only unavoidable hard costs. The rest is scoped to how polished the handover needs to be.

06 Read this before you hand it over — the honesty section

Your own deployment runbook (docs/DEPLOYMENT.md §8) lists seven known gaps. That candour is a strength — but an enterprise infosec team *will* find these, so decide deliberately which to close before handover. This is the "don't look stupid" list, ranked by how much it matters for a TATA/L&T/TVS-grade client holding real employee PII.

CLOSE FIRST

No DPDP / GDPR data-subject flows

part of hardening
engineering

No "export my data" or "delete my account". Deleting a user who submitted ideas only deactivates them. Under India's DPDP Act this is a real compliance gap when the tenant is a large regulated enterprise.

CLOSE FIRST

Uploads validated by extension, not content

ClamAV

~₹0 sw + setup

A file renamed to .pdf is accepted (it's stored inert and served as an attachment, so it can't execute — but it isn't virus-scanned). Put ClamAV in front before staff open these routinely.

BEFORE SCALE-OUT

Rate limits are per-process, in memory

config change

Redis budgeted

The moment you run more than one Node instance (the Production tier does), the effective limit multiplies. Move to the Redis-backed store — the Redis line is already in the Production budget.

RECOMMENDED

JWT stored in localStorage

engineering

next iteration

Any XSS in the React app can read the token. The strict CSP in the runbook is the current mitigation; an httpOnly cookie + CSRF is the stronger next step. Infosec reviewers routinely flag this.

RECOMMENDED

No automated test suite

engineering

ongoing

Every fix so far was verified by hand. With no tests, a regression won't be caught automatically — the biggest long-term maintenance risk, and something enterprise buyers ask about.

RECOMMENDED

No tamper-evident security audit log

engineering

next iteration

Logins, lockouts and password resets go to the app log, not a tamper-evident store. (The *workflow* audit trail — who approved what — is complete and in the DB.)

- ✓ Already handled well and worth saying out loud: the server **refuses to boot** on insecure config, uses a least-privilege DB user, forces HTTPS/HSTS, enforces a forced first-password change, isolates every tenant in its own schema, and streams the bulk importer so a hostile file can't exhaust memory. This is a genuinely well-hardened codebase — the gaps above are the *remaining* 10%, not the whole picture.

07 Bottom line — what to put in front of IFQM

LINE	RECOMMENDATION	COST
Where it runs	Cloud VM in an India region + managed MySQL (AWS/Azure Mumbai for enterprise clients)	—
Frontend	Vercel / Cloudflare Pages (static) or same Nginx server	Free
Monthly hosting — pilot	Single tenant, proof of value	₹7k–9k / mo
Monthly hosting — production	Recommended · multi-tenant, HA, no-lag	₹28k–38k / mo
Monthly hosting — enterprise HA	Multi-AZ AWS/Azure, WAF, compliance-grade	₹90k–1.5L / mo
Gemini AI scoring	Pay-as-you-go ceiling	₹200–1,500 / mo
Domain	.com / .in, renews yearly	₹800–1,500 / yr
SSL	Let's Encrypt / ACM	Free
One-time hardening + handover	Close \$6 gaps + deploy + (optional) VAPT	₹2L–5L
Recommended go-live	Production tier + hardening, first year	≈ ₹6L–9L yr 1

Year-1 figure = ~₹4L hosting (₹33k avg × 12) + ~₹2–5L one-time hardening & deployment. From year 2 it settles to roughly **₹4L–4.5L / year** in pure hosting for the Production tier, before any per-tenant pricing you charge IFQM on top. If tenants scale past ~10 large orgs, step up compute and the DB tier and re-cost — the architecture scales horizontally on the app tier and vertically on MySQL.

Assumptions & caveats. Prices are provider list rates as of mid-2026, converted at ₹85/USD, before reserved-instance or committed-use discounts (which typically save 20–40% on compute). "Users per tenant" is the employee register, not peak concurrency; sizing assumes 1–5% concurrent, typical for an ideation tool. Managed-MySQL, Redis, email and CDN pricing vary by provider (DigitalOcean/AWS/Azure) — treat tier totals as ±15%.

Verify before quoting. Confirm current provider pricing and the client's mandated cloud/region before you commit numbers in writing — large enterprises often dictate AWS or Azure and an India region for data residency, which sets the tier for you.

Prepared for internal planning · IFQM Ideation Feature Quality Matrix · figures are estimates, not a fixed quotation.